THEJUS MAHAJAN PH.D.

Computational Biologist — AI & Multi-Omics

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*30.09.1992 (Talikulam, India)
Married

PROFILE

- **Focus:** AI-driven multi-omics integration, deep learning, and bioinformatics pipeline development for translational biomedicine.
- **Core Stack:** Python (JAX, PyTorch concepts, Scikit-learn, Pandas), R (Bioconductor, Tidymodels), Nextflow (DSL2), BASH.
- **Current Objective:** Building AI frameworks for multi-omics data integration to advance research in stem cell biology and complex comorbidity networks.

EXPERIENCE

Internship – Bioinformatics & Data Pipeline Engineering

HealthTwiSt GmbH

02/2026 - 04/2026 | Berlin, Germany

Clinical data research and development under Dr. Andreas Busjahn

- Refactored a monolithic 1,834-line R data pipeline (tidyverse) for Germany's interventional radiology quality registry (DeGIR); 143,000+ patient records from ~300 clinics. Reduced to 1,348 lines (−26.5%) with byte-identical output verified via `identical()`.
- Externalised 257 hard-coded correction rules into 8 configuration CSV files, enabling non-programmer editing of business rules.
- Replaced 357 deprecated function calls across 4 pipeline files, modernising the codebase to current tidyverse standards.
- Created the `degirtools` R package (9 exported functions, roxygen2 docs, `devtools::check()` passing) by extracting reusable logic from a 767-line helper file.
- Discovered 2 pre-existing bugs through systematic code analysis; documented both and preserved original behaviour for supervisor review — demonstrating production-code discipline.
- Built an interactive R Shiny dashboard (6 modules: overview, interventions, complications, radiation doses, success rates, about) using plotly, DT, and bslib. Deployed to shinyapps.io with GDPR-safe synthetic data.
- Wrote Tidymodels teaching materials (Quarto/GitHub) replacing the legacy caret framework for ML workflows in R.

Further Training – Bioinformatics and Biostatistics

CQ Beratung + Bildung GmbH

08/2025 - 02/2026 | Berlin, Germany

Application-related Bioinformatics and Biostatistics

- Programming methodology: Shell (BASH), Python (object-oriented), R.
- Bioinformatics databases (NCBI, Biopython, SQL) and sequence analysis.
- NGS analysis pipelines (Nextflow, Galaxy); structural bioinformatics.
- Clinical biostatistics (R/Bioconductor, ANOVA, PCA, HCA, survival analysis); GDPR/BDSG compliance.
- Capstone: Built a reproducible Nextflow (DSL2) pipeline for Hepatitis Delta Virus sequence analysis — automated NCBI download, MAFFT alignment, and trimAl trimming with Conda-managed dependencies (GitHub).

Guest Scientist

Helmholtz-Zentrum Hereon, Ökosystemmodellierung

05/2025 - 10/2025 | Geesthacht, Germany

Computational Biologist & Systems Modeler

- Spearheaded research modeling evolutionary processes in marine ecosystems.

STRENGTHS

Data Pipeline Engineering
Refactored a production data pipeline (1,834 lines, 143K records, ~300 clinics) with 26.5% code reduction. Built degirtools R package. Byte-level verification after every change via `identical()`.

Complex Problem-Solving
Resolved a critical data-loss error by creating a Fortran simulation to correct for a misplaced particle detector, recalculating results using multi-sensor data.

AI & Multi-Omics Integration
Deep expertise in tensor operations and GPU/TPU-accelerated computing (Google JAX) — the computational foundation for graph neural networks and variational autoencoders. Combines statistical genomics (R/Bioconductor) with Python-based deep learning to integrate high-dimensional biological datasets.

TECHNICAL SKILLS

Programming
Python (JAX, Pandas, NumPy, Scikit-learn, Biopython), R (Bioconductor, Tidymodels), SQL, BASH

Bioinformatics
Multi-Omics Data Integration, Nextflow (DSL2), NGS Analysis, Multiple Sequence Alignment (MAFFT), Sequence Trimming (trimAl), NCBI / Entrez

AI & Deep Learning
Google JAX (TPU/GPU), PyTorch & TensorFlow concepts, Graph Neural Networks (conceptual), Scikit-learn, Tidymodels, PCA, Clustering

Data Engineering
Hardware Acceleration (TPU/GPU), Large dataset processing (HDF5, NetCDF), Pipeline development, R Shiny, devtools/roxygen2

Tools
Linux/Unix, Git, HPC clusters, Conda

TRAINING / COURSES

IBM: Machine Learning with Python; Introduction to AI
Coursera (in progress)

High Performance Computing (HPC) Training
Jülich Supercomputing **Center (JSC)**,

- Applied advanced computational techniques to simulate long-term environmental impacts.
- Finalised the computational modelling framework and prepared the simulation data for peer-reviewed publication.

Job Search & German Language Learning
Hamburg, Germany
02/2025 - 04/2025 · Hamburg, Germany

Active job search and German language acquisition (Goethe B1 certification).

Post-doctoral Scientist
University of Hamburg, Institute of Marine Ecosystem and Fisheries Sciences
08/2021 - 01/2025 · Hamburg, Germany

Marine Ecosystem Modelling
- *Elternzeit (Parental Leave) from 01/2024 to 12/2024.*
- Model Development: Developed the novel "Cyanobacteria Life Cycle" (CLC) model within the ERGOM framework for climate warming projections.
- High-Performance Computing & ML: Led the translation of a legacy Fortran continuous model into a high-performance Python simulation engine. Utilized **Google JAX** vectorization and branchless logic for massive parallelization (direct compatibility with **TPUs and GPUs**).
- This architecture established a strong foundation in tensor operations underpinning deep learning frameworks like **PyTorch** and **TensorFlow**.

Job Search
Kerala, India
04/2021 - 07/2021 · Kerala, India

Relocation from India to Germany and active job search.

Physics Educator
Tutorwaves Solutions Inc / Freelance
10/2018 - 03/2021 · Kerala, India

Prepared physics content for competitive examinations and coached state-level chess teams.

Ph.D. Researcher in Astrochemistry
University of Paris-Saclay
10/2015 - 09/2018 · Orsay, France

Led atomic collision experiments, analyzed data, and modeled outcomes.
- Acquired and processed terabytes of raw data from tandem particle accelerator experiments.
- C++ signal processing, algorithm development, and molecular fragmentation modeling (KIDA database).

Forschungszentrum Jülich

MITx: 7.00x Introduction to Biology - The Secret of Life
Comprehensive coursework on genetics, biochemistry, and molecular biology.

OTHER SELECTED PROJECTS

Hepatitis Delta Virus Pipeline • GitHub
Nextflow (DSL2) pipeline for viral sequence analysis: NCBI Entrez download, MAFFT alignment, trimAl trimming, Conda-managed dependencies.

JAX/TPU High-Performance Simulator • GitHub
Ported a legacy Fortran model to a modular Python architecture using Google JAX. Engineered branchless, vectorized logic for massive TPU/GPU parallelization.

Introduction to Tidymodels • GitHub
Teaching script for ML in R: recipes, parsnip, workflows, and yardstick on the Palmer Penguins dataset.

Berlin Tree Analysis • GitHub
Interactive Jupyter notebook analyzing 5 years of tree planting data in Berlin using Python, Pandas, and Seaborn.

PUBLICATIONS

5 peer-reviewed articles • Google Scholar Profile

LANGUAGES

English ●●●●○
Proficient

German ●●●○○
B1 - Goethe Certified

French ●●○○○
Intermediate

Malayalam ●●●●●
Native

Tamil ●●●●○
Proficient

Hindi ●●○○○
Intermediate

EDUCATION

Ph.D. in Astrochemistry
Université Paris-Saclay, Institute of Molecular Sciences (ISMO)
10/2015 - 09/2018 · Orsay, France

Supervisor: Dr. Karine Béroff. Thesis: *Excitation and fragmentation of C_nN^+ (n=1–3) in collisions with He atoms at intermediate velocity.*

M.Sc. in Physics
National Institute of Technology Calicut, India
07/2012 - 12/2014 · Calicut, India

First Class with Distinction (CGPA 8.71/10).

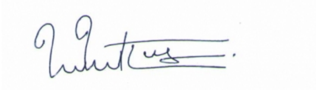
Research Project – Theoretical Atomic Physics
IIT Mandi (under Dr. Hari Varma)
02/2015 - 08/2015 · Mandi, India

Theoretical atomic physics; partly remote.

B.Sc. in Physics
University of Calicut, India
06/2009 - 04/2012 · Thrissur, India

Grade: B+ (CGPA 3.49/4.0).

Hamburg, 11 April 2026



Dr. Thejus Mahajan